YEAR 11 PHYSICS, AQA GCSE, LESSON 4 OF 10

# Atoms & Nuclear Radiation.

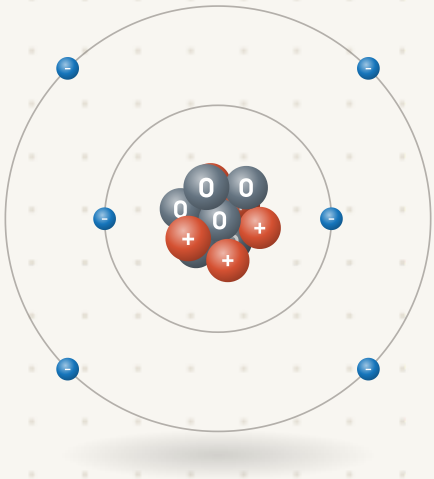
## Lesson 04 Energy levels and the nucleus

Neon signs glow red and sodium street lamps glow orange. Each element gives out its own colours of light. What are its electrons doing?

IN THIS LESSON YOU WILL

- describe Bohr's model of electrons in energy levels
- explain how electrons change level when they absorb or emit electromagnetic radiation
- describe how protons and then neutrons were discovered
- explain why scientific models change

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

# How this booklet works.

This is the booklet for **Lesson 4** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

**DO NOW**
Quick questions from earlier lessons. Remembering again is what makes it stick.

**LEARN IT**
Explanations, diagrams and worked examples. Read this again when you revise.

**TRY IT**
Practice that builds up: first with support, then on your own, then a challenge.

**LAB**
Record what you find in a simulation or a model experiment.

**EXAM CORNER**
AQA-style questions with a tip on how the marks are given.

**REVISE IT**
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

## The route through the unit (this lesson is highlighted)


01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

**04 Energy levels and the nucleus**
The Bohr model, absorbing and emitting EM radiation, protons and neutrons, why models change (slides 18–25)

05 Radioactive decay

06 Nuclear equations

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation


Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

LESSON

# 04 Energy levels and the nucleus.

AQA Physics 4.4.1.1–4.4.1.3; Combined Science 6.4.1.1–6.4.1.3

BIG QUESTION

**Neon signs glow red and sodium street lamps glow orange. Each element gives out its own colours of light. What are its electrons doing?**

BY THE END YOU CAN

- describe Bohr's model of electrons in energy levels
- explain how electrons change level when they absorb or emit electromagnetic radiation
- describe how protons and then neutrons were discovered
- explain why scientific models change

Bohr model · energy level · absorb · emit · electromagnetic radiation · proton · neutron · peer review

SLIDES AND SIMULATIONS

**Teaching deck, slides 18–25**

The Bohr model, energy levels, protons then neutrons, and why models change.

panphy.app/gcsephy/decks/atoms-and-radiation/#18

**Atomic Models simulation**

panphy.app/simulations/atomic_models.html

RETRIEVE

## Do now

From earlier lessons. No notes!

1 Which experiment showed that atoms have a tiny nucleus?

The alpha scattering experiment.

2 What is the charge on an alpha particle?

Positive (+2).

3 In the nuclear model, where are the electrons?

Outside the nucleus.

EXPLAIN

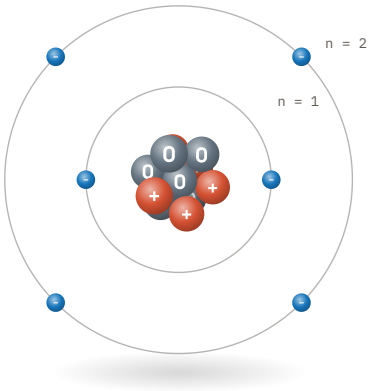
# Filling in the details

## Bohr's model (1913)

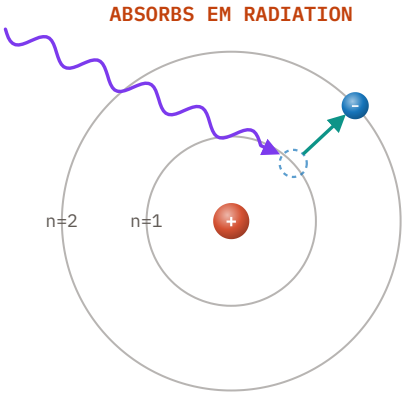
Niels Bohr adapted the nuclear model. He suggested that electrons orbit the nucleus **at specific distances**. Each allowed distance is an **energy level**.

Bohr's theoretical calculations **agreed with experimental observations** of the light given out by atoms. That agreement is why other scientists accepted his model.

A carbon atom has 2 electrons in the first energy level (n = 1) and 4 in the second (n = 2).

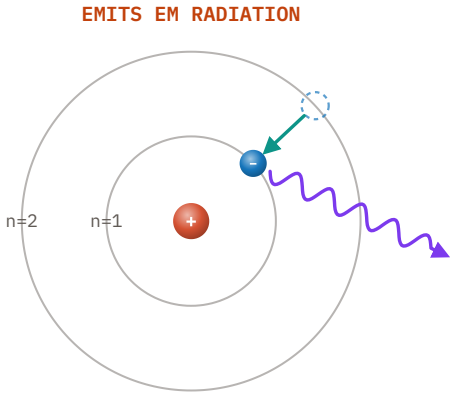



## Electrons can change energy level




**moves further out**

higher energy level




**moves closer in**

lower energy level

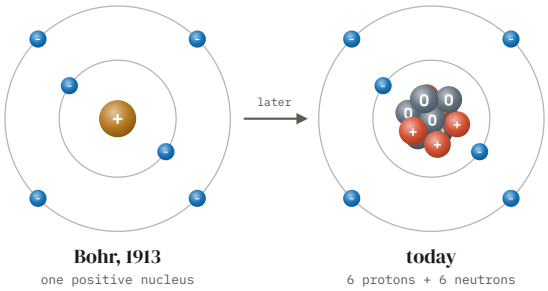
| The electron… | Moves… | Energy level |
|---|---|---|
| **absorbs electromagnetic (EM) radiation** | **further from** the nucleus | higher |
| **emits electromagnetic (EM) radiation** | **closer to** the nucleus | lower |

THINK OF A LADDER

An electron can stand on a rung, but not between rungs. To climb up it must take in exactly the right amount of energy; when it drops down, it gives that energy out as EM radiation. This is why each element gives out its own colours of light.

**Where the model breaks:** the energy levels are not evenly spaced like rungs, and electrons are not little balls standing on them.

## Filling in the nucleus



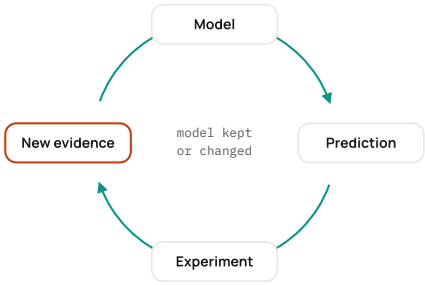

Bohr pictured one positive nucleus. Today we draw its protons and neutrons.

Later experiments (1919) showed that the positive charge of a nucleus is made of smaller particles, each with the same positive charge: **protons**.

In 1932 **James Chadwick** gave the experimental evidence for the **neutron**, about 20 years after scientists had accepted that atoms have a nucleus.

## Why models change

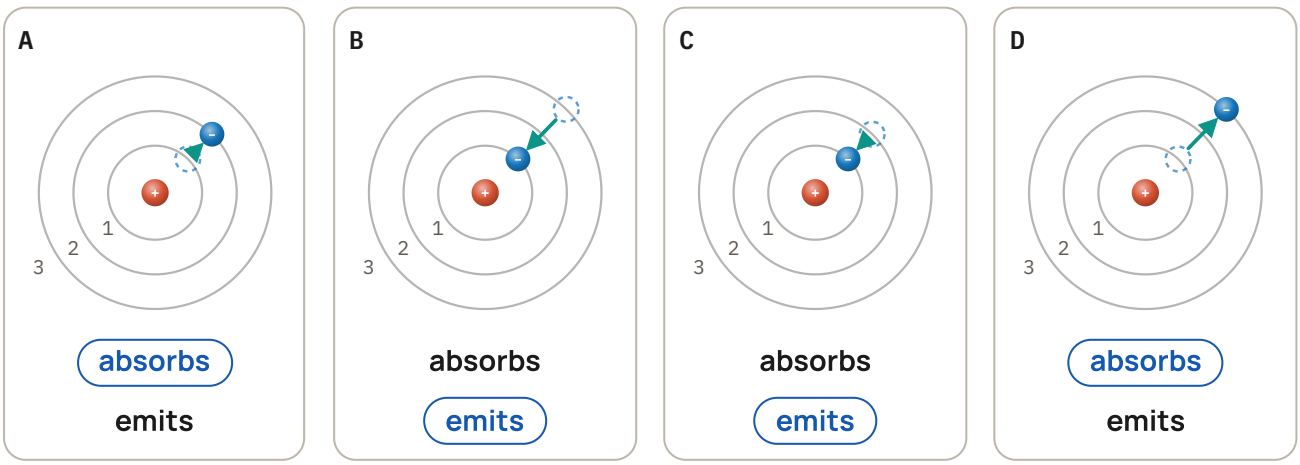
1. A model must **explain the evidence** we already have.
2. It makes **predictions** that experiments can test.
3. If new evidence does not fit, the model is **changed or replaced**.
4. Results are **published** so other scientists can check them (**peer review**) before a model is accepted.




PRACTISE

## Try it

**1** In each diagram an electron moves between energy levels. Does the atom **absorb** or **emit** EM radiation? Circle one for each.

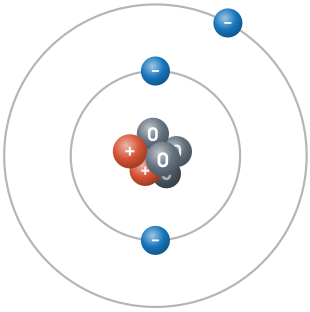


**2** Complete the sentences.

When an electron absorbs electromagnetic radiation, it moves to a **higher** energy level, **further from** the nucleus. When an electron moves to a lower energy level, it **emits** electromagnetic radiation.

WORD BANK: higher | lower | further from | closer to | absorbs | emits

**3** A lithium atom has 3 electrons: 2 in the first energy level and 1 in the second. Draw the electrons on the diagram.



**4** SUPPORTED Complete the timeline.

| Year | Scientist(s) | Discovery or model |
|---|---|---|
| 1897 | J. J. Thomson | The electron (then the plum pudding model, 1904) |
| 1909–1911 | Rutherford, Geiger and Marsden | Alpha scattering; the nuclear model |
| 1913 | Niels Bohr | Electrons in energy levels |
| 1932 | James Chadwick | Evidence for the neutron |

**5** Explain why scientists really change models.

**Leo**: Scientists keep changing the model of the atom because they get bored of the old one.

When new evidence (e.g. from an experiment) does not fit the old model, the model has to be changed or replaced so it explains all the evidence. The new model is checked by other scientists (peer review) before it is accepted.

6 CHALLENGE Protons were identified in 1919, but the neutron was not found until 1932. Suggest why the neutron was harder to find.

A neutron has no charge, so it is not deflected by electric or magnetic fields and does not ionise atoms directly, so it does not show up easily in detectors.

AQA-STYLE

## Exam corner

Write in full sentences. Show every step of a calculation.

**Get the marks:** Use the key words: absorb/emit, higher/lower energy level, further from/closer to the nucleus.

7 Describe how the arrangement of the electrons in an atom may change when the atom absorbs electromagnetic radiation. [2 marks]

An electron moves to a higher energy level (1), further from the nucleus (1).

8 Explain why Bohr's model of the atom was accepted by other scientists. [2 marks]

His theoretical calculations agreed with experimental observations (1); it could be tested and checked by other scientists / explained evidence the nuclear model could not (1).

9 Give the name of the scientist who provided the evidence for neutrons. [1 mark]

James Chadwick.

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REVIEW LATER

# Revise it

Use this box after the lesson and before tests.

## SUMMARY

- Bohr (1913): electrons orbit at specific distances, called energy levels. His calculations agreed with experiments.
- Absorb EM radiation → electron moves to a higher level, further from the nucleus.
- Emit EM radiation → electron moves to a lower level, closer to the nucleus.
- Protons were identified later (1919); James Chadwick gave evidence for neutrons (1932).
- New evidence changes models; results are published and peer reviewed.

## COVER THE PAGE AND ANSWER

**1** What did Bohr add to the nuclear model?

**2** What happens when an electron emits EM radiation?

**3** Who gave the evidence for the neutron?

**4** Why are scientific results published?

## I CAN…

- describe Bohr's model of the atom
- explain what happens when electrons absorb or emit EM radiation
- describe the discovery of protons and neutrons
- explain why models of the atom have changed

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REFERENCE

# Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

## The three particles

Lesson 1

| Particle | Charge | Mass | Where |
|---|---|---|---|
| **Proton** | +1 | 1 | nucleus |
| **Neutron** | 0 | 1 | nucleus |
| **Electron** | −1 | very small | energy levels |

## Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ $1 \times 10^{-10}$ m; nucleus radius < 1/10 000 of that

## Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

## Alpha scattering

Lesson 3

| Observation | Conclusion |
|---|---|
| **most straight through** | mostly empty space |
| **some deflected** | positive centre |
| **a very few bounced back** | mass in a tiny centre |

## Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

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REFERENCE

# Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

**Absorb**: Take in energy. An electron that absorbs EM radiation moves to a higher energy level.

**Bohr model**: Electrons orbit the nucleus at specific distances, called energy levels (1913).

**Emit**: Give out. An electron that moves to a lower energy level emits EM radiation.

**Energy level**: A specific distance from the nucleus at which electrons are found.

**Peer review**: Checking of published findings by other scientists.

CHECK YOURSELF

# Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

### DO NOW

1. The alpha scattering experiment.
2. Positive (+2).
3. Outside the nucleus.

### COVER THE PAGE

1. Electrons orbit at specific distances (energy levels).
2. It moves to a lower energy level, closer to the nucleus.
3. James Chadwick (1932).
4. So other scientists can check them (peer review).